The derivative singlet at first order
Fixed-time connected four-point dynamics

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Abstract

We apply established semiclassical chord four-point functions to a derivative singlet of complex DSSYK after removing its conserved part. Keeping all three same-species pairings gives an explicit connected kernel at arbitrary fixed real times, and closed first-order expressions for the commutator squared and the adjacent-time cumulant. Exact chord moments and a finite-deformation four-arc implementation provide independent checks. This supplies a microscopic interaction target for the Susskind Rindler-meson program.

1 Observable, state, and result

We use the full annealed normalized trace τ at $\beta = \mu = 0$, standard CAR, and units $\hbar = c_{\text{light}} = 1$. In the normalization of the preceding local singlet notes,

$$\begin{aligned} A &= \sum_i [H, c_i]^\dagger [H, c_i], & O &= A - \mathbb{E}A(Q), \\ B &= H^2 - \mathbb{E}H^2(Q), & X &= (O - cB)/\sqrt{W_d}, \\ c &= 2\lambda \frac{1-q}{1+q}, & W_d &= 4\mathcal{J}_s^4 \left[\lambda^{-1} + \frac{4q}{1+q}\right]. \end{aligned} \tag{1}$$

Here Q is fermion number. The Hermitian Gaussian ensemble is $H = \sum_{|I|=|J|=p} G_{IJ} c_I^\dagger c_J$, including overlaps, with

$$\mathbb{E}G_{IJ}G_{KL} = w\delta_{IL}\delta_{JK}, \qquad w = 4^p \frac{\mathcal{J}_s^2}{\lambda} \left(\frac{p!}{N^p}\right)^2.$$

First take $N, p \to \infty$ at $p^2/N = \lambda > 0$; then expand at small λ with fixed string times. Our previous mixed-chord reduction has $Q_{HH} = q$, $Q_{HX} = q^2$, $Q_{XX} = q^4$. Set

$$h = 4\lambda, \quad q = e^{-h}, \quad H_{\text{src}} = H/(2\mathcal{J}_s), \quad u_j = 2\mathcal{J}_s t_j, \quad d = 2, \quad \tau(H_{\text{src}}^2) = 1/h. \tag{2}$$

Write $g_{ij} = \operatorname{sech}^{2d}(u_i - u_j)$ and $T_{ij} = \tanh(u_i - u_j)$. The cumulant subtracts the *full finite-h* two-point functions:

$$\kappa_4 = \tau(X_1X_2X_3X_4) - G_{12,h}G_{34,h} - G_{23,h}G_{41,h} - G_{13,h}G_{24,h}. \tag{3}$$

The resulting fixed-time semiclassical coefficient is

$$\begin{aligned} \kappa_4 &= hK_4 + O(h^2), \\ K_4 &= -d^2 \left[g_{12}g_{34}T_{12}T_{34} + g_{23}g_{41}T_{23}T_{41} \right. \\ &\qquad \left. + g_{13}g_{24}\frac{\cosh(u_1 + u_3 - u_2 - u_4)}{\cosh(u_1 - u_3)\cosh(u_2 - u_4)}\right]. \end{aligned} \tag{4}$$

The formula uses the sourced expansion in Refs. [1, 2]; the remainder notation is asymptotic, not a certified uniform numerical bound.

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2 Two useful closed functions

Define $X(t) = e^{iHt}Xe^{-iHt}$ and the positive tracial observable

$$\mathcal{C}(t) = \tau([X(t), X]^\dagger[X(t), X]) = 2\{F_{22}(t) - F_{\rm OTO}(t)\}, \tag{5}$$

where $F_{22} = \tau(X(t)^2X^2)$ and $F_{\rm OTO} = \tau(X(t)XX(t)X)$. At both orderings the disconnected part is $1 + 2G_h(t)^2$. It cancels exactly before any expansion. With $u = 2\mathcal{J}_s t$, Eq. (4) gives

$$K_{22}(u) = d^2(2\tanh^2 u - 1)\,\mathrm{sech}^{4d} u, \tag{6}$$

$$K_{\rm OTO}(u) = -d^2\left[\cosh(2u) + 2\tanh^2 u\ \mathrm{sech}^{4d} u\right], \tag{7}$$

$$\mathcal{C}(t) = 2hd^2\left[\cosh(2u) + (4\tanh^2 u - 1)\,\mathrm{sech}^{4d} u\right] + O(h^2). \tag{8}$$

For our $d = 2$ singlet these become

$$\boxed{\mathcal{C}(t) = 32\lambda\left[\cosh(4\mathcal{J}_s t) + (4\tanh^2(2\mathcal{J}_s t) - 1)\,\mathrm{sech}^8(2\mathcal{J}_s t)\right] + O(\lambda^2),} \tag{9}$$

$$\kappa_{22}(t) = 16\lambda(2\tanh^2(2\mathcal{J}_s t) - 1)\,\mathrm{sech}^8(2\mathcal{J}_s t) + O(\lambda^2). \tag{10}$$

The box in Eq. (9) concerns the model coefficient, with the limits and normalization specified above. It is not an on-shell bulk coupling.

At short times, the coefficient reproduces

$$\mathcal{C}(t) = 1280\mathcal{J}_s^2\lambda t^2 + O(\lambda t^4\mathcal{J}_s^4) + O(\lambda^2), \qquad \kappa_{22}(0) = -16\lambda + O(\lambda^2). \tag{11}$$

Also $\kappa_{22}''(0) = 768\mathcal{J}_s^2\lambda + O(\lambda^2)$. The leading adjacent cumulant changes sign at

$$\mathcal{J}_s t = \tfrac{1}{2}\,\mathrm{artanh}(1/\sqrt{2}) = 0.4406868\ldots.$$

This is a zero of the coefficient, not an assertion about the exact finite-λ zero. The first-order commutator coefficient is positive for $t \neq 0$: if $\tanh^2 u \geq 1/4$ both terms in brackets are nonnegative; otherwise their sum exceeds $1 - \mathrm{sech}^8 u > 0$.

For $1 \ll \mathcal{J}_s t$ within a small-correction overlap window, Eq. (9) has leading behavior $16\lambda e^{4\mathcal{J}_s t}$. This matches the earlier scrambling expansion. Taking times of order $\log(1/\lambda)/(4\mathcal{J}_s)$ requires the resummed kernel; the fixed-time expansion alone does not justify that limit.

3 What cancels and what must remain

An expansion of G_{q^2} or G_{q^4} has its own first-order correction. Those terms must be included if one computes F_{22} in isolation. Working with connected channels removes them algebraically. It does *not* remove either of the uncrossed channels. Retaining only the crossed contribution would fail the adjacent cumulant and the early-time commutator checks.

4 Source map and oriented insertions

The tools are the two-sided OPE of Lin–Stanford, including their Eq. (205) and Appendix G, and Streicher's time-ordered and OTO correlators [1, 2]. Their chord deformation parameter is our $h = 4\lambda$. Streicher's interaction-order symbol q is not our chord weight $q = e^{-h}$.

For the crossed pairing (13)(24), continuation of Lin–Stanford's two-sided variables at zero microscopic inverse temperature gives

$$\tau_+ = i(u_3 - u_1), \quad \tau'_+ = i(u_4 - u_2), \quad \tau_- = i(u_1 + u_3 - u_2 - u_4), \quad c_{\mathrm{LS}} \to 1. \tag{12}$$

The surviving normalized connected term in their (205) is $-hd^2 \cos(\tau_-)/[\cos(\tau_+)\cos(\tau'_+)]$. Multiplication by $g_{13}g_{24}$ produces the last term of Eq. (4). At coincident times it equals $-hd^2$. The mutual crossing factor q^{d^2} is already represented in this connected result; multiplying it a second time would double count.

For an uncrossed pairing, Appendix G.2 identifies leading energy exchange. At zero inverse temperature the two-energy measure is invariant under exchange of its arguments and simultaneous sign reversal. Thus, with $s = u_i - u_j$ and the indicated operator order,

$$\tau(H_{\mathrm{src}} X(u_i) X(u_j)) = -\frac{i}{2}\partial_s G_h(s) = id \tanh(s)\,\mathrm{sech}^{2d} s + O(h). \tag{13}$$

Dividing the product of two such matrix elements by $\tau(H_{\mathrm{src}}^2) = 1/h$ yields the first term of Eq. (4). Cyclic rotation yields the second, with *oriented* separation $u_4 - u_1$. Replacing T_{41} by T_{14} would reverse that contribution. In Appendix G.1 one must take the zero-inverse-temperature limit after forming the products: a vanishing variance prefactor multiplies divergent tangent coefficients.

5 An exact finite-q computational representation

We generalize the previous BINT transfer implementation [3] to four independent real times. In $\mathcal{J}_s = 1$ units let

$$J_{n,n+1} = \lambda^{-1/2}\sqrt{\frac{1-q^{n+1}}{1-q}}, \quad D_r = \mathrm{diag}(r^n), \quad r = q^2, \quad \chi = q^4, \tag{14}$$

and set $a = (t_4 - t_3, t_3 - t_2, t_2 - t_1, t_1 - t_4)$, $U_j = e^{ia_j J}$. The two uncrossed contractions are

$$A_0 = (D_r U_3|0\rangle)^T U_2 U_4 (D_r U_1|0\rangle), \qquad A_1 = \text{one cyclic shift of the four arc labels.} \tag{15}$$

The transpose is bilinear, without complex conjugation. This distinction also matters for the general crossed contraction on the next page.

Put $f_n = (q;q)_n$, $b_i = (r^2;q)_i r^i/f_i$, and define the source matrix

$$V_{li}^{(j)} = \sqrt{b_i}\, r^l \sqrt{f_{l+i}/f_l}\, (U_j)_{l+i,0}. \tag{16}$$

The remaining channel and full function are

$$R = \sum_{n,i} r^n (U_3 V^{(4)})_{ni} (U_2 V^{(1)})_{ni}, \qquad F = A_0 + A_1 + \chi R. \tag{17}$$

Only for the alternating ordering does R reduce to the previous positive Hilbert–Schmidt norm. General real times require the bilinear product in Eq. (17).

6 Evidence and measured finite-deformation errors

Exact integer chord polynomials, expanded with rational arithmetic in $q = e^{-h}$, independently match 22 Taylor coefficients of Eq. (4). The checks include 72 cancellations of nonpositive Laurent powers. The time rays $(1,0,1,0)$ and $(1,1,0,0)$ extend through u^{10}; $(2,1,-1,0)$ and $(0,1,3,-1)$ extend through u^8. For example,

$$\frac{\mathcal{C}}{h} = 80u^2 - \frac{640}{3}u^4 + \frac{4352}{9}u^6 - \frac{50888}{63}u^8 + \frac{3150752}{2835}u^{10} + O(u^{12}). \tag{18}$$

These checks constrain the closed formula locally, rather than proving its remainder for arbitrary time.

A second implementation evaluates the original nonsymmetric BINT index sum at 50 decimal digits, at the same finite size $L = 14$. All 12 channel comparisons at two deformations and two nonalternating time sets agree with the symmetric implementation to absolute tolerance 10^{-12}. For the main computation, $\lambda = (.04, .02, .01, .005)$ uses respectively $L = (169, 332, 672, 1383)$. Selected $L+16$ channel changes are at most 1.23×10^{-15}. Alternating reduction, cyclicity, reversal and common time translation also pass. These are numerical representation and stability checks, not a bound on all floating-point error.

$\mathcal{J}_s t$	predicted $\mathcal{C}/h$	extrapolated $\mathcal{C}/h$	relative difference
0.10	2.887655	2.869919	−0.61%
0.20	8.887575	8.819786	−0.76%
0.35	17.805892	17.465716	−1.91%
0.50	30.426106	28.771611	−5.44%

Here extrapolation means $2(\mathcal{C}/h)_{\lambda=.005} - (\mathcal{C}/h)_{\lambda=.01}$; it is not an error certificate. At $\mathcal{J}_s t = .5$, the raw $\lambda = .005$ coefficient is 24.850065, or 18.33% below the leading prediction. The corresponding extrapolated adjacent coefficient is 0.01986431 against 0.01991637, a 0.26% difference.

The previous alternating-kernel cutoff parameters were reused conservatively. Cauchy–Schwarz bounds a general bilinear contraction by the product of its two Hilbert–Schmidt norms, but we have not completed a new total cutoff proof for every arc here. The saved alternating cutoff bound is therefore not a certified general-time error bar; selected refinements provide the empirical control stated above.

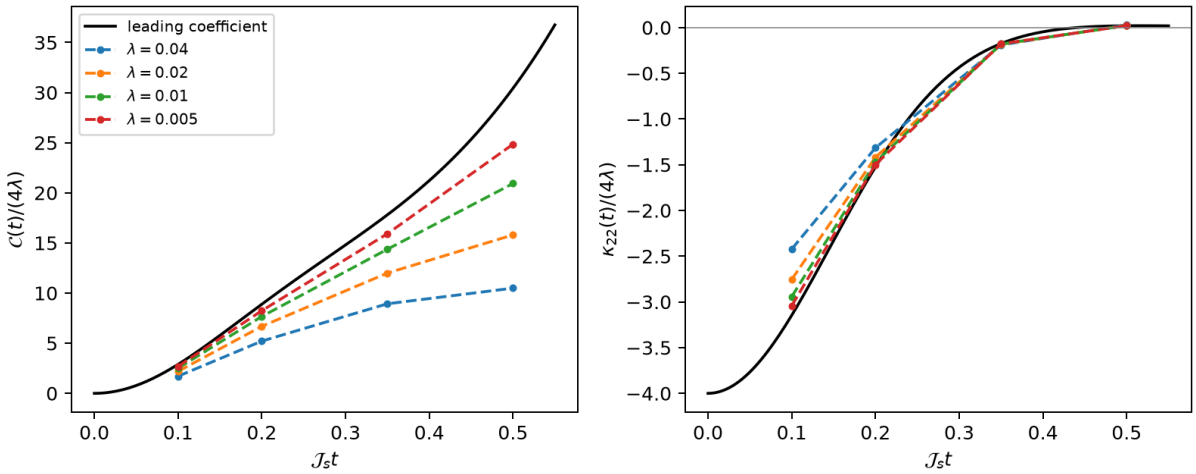


Figure 1: Finite-λ kernels compared with their first-order coefficients. The commutator curve approaches its limit more slowly at the later times.

For the three nonalternating physical time sets

$$(.3, .1, -.1, 0), \qquad (.2, -.05, .35, .1), \qquad (.15, .15, -.15, -.15),$$

the errors in κ_4/h at $\lambda = .005$ are respectively -0.0038913, $+0.2446429$, -0.0014850. All shrink from .01 to .005; the third is not monotone over the entire four-deformation scan. Maximum observed crossed-sum cancellation is 2.20, and connected-subtraction condition number is 373. Absolute and coefficient errors are retained in the machine-readable data.

7 The next physical construction

The earlier regulated KG packet matched the singlet's midpoint two-point function, with arbitrary positive mass absorbed in radial dilation. Its ordinary equal-thermal-gap four-point value 33 at zero real time does not equal the microscopic value 3. The present calculation fixes the *microscopic* three-channel multi-insertion target, using the pair-dependent continuation underlying the fake-circle construction. It has not yet supplied a spatial KG operator map preserving multiplication.

The next task is to construct an independent spatial or translation observable and its cross-correlators with X. We will first examine the chord generators and multiparticle representation, then test whether a Rindler translation algebra and a mass observable survive the specified limit. The chord label $d = 2$ alone does not determine a Poincare mass. This moves the project from temporal kernel matching toward the meson spectrum and physical scattering dictionary.

References

[1] H. W. Lin and D. Stanford, *A symmetry algebra in double-scaled SYK*, arXiv:2307.15725.

[2] A. Streicher, *SYK Correlators for All Energies*, arXiv:1911.10171.

[3] M. Berkooz, M. Isachenkov, V. Narovlansky and G. Torrents, *Towards a full solution of the large N double-scaled SYK model*, arXiv:1811.02584.